Semantic Theory 2026: Exercise 8 Key

Question 1

Consider the following sentences:

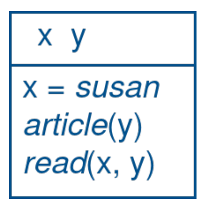
i. *Susan reads an article.*

ii. *The team does not win every match.*

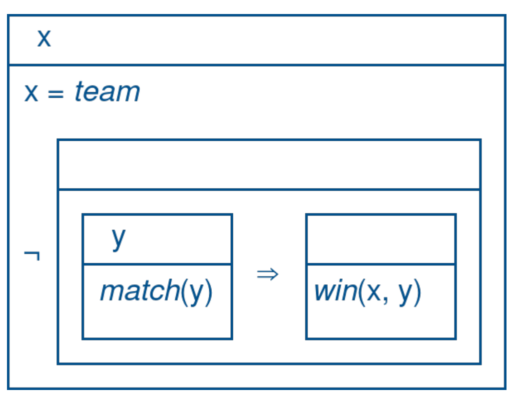
iii. *If someone goes to a casino, they will lose money.*

a. Give DRS representations for each of these sentences. Underlined terms can be treated as named entities.

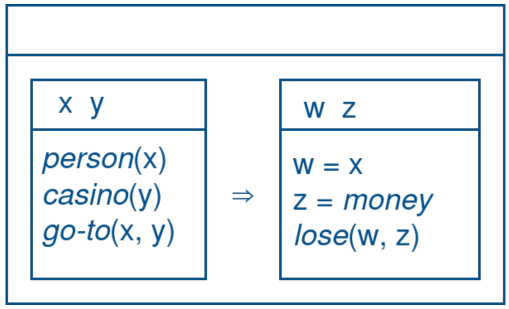
i.



ii.



iii.



b. Determine for each DRS which discourse referents are available for anaphoric reference (i.e. from a subsequent sentence).

i. $\{x, y\}$

ii. $\{x\}$

iii. $\varnothing$

c. Give the truth-conditions for the DRS corresponding to (iii). Use the verifying embeddings to arrive at the model-theoretic interpretation.

The DRS $K = (U_K, C_K)$ is true in a model $M = (U_M, V_M)$ iff there exists $f\colon U_D \nrightarrow U_M$ such that:

1. $U_K \subseteq \mathit{dom}(f)$: $U_K = \varnothing$, so this condition is always (trivially) satisfied.
2. f verifies K in M: $f \models_M K$

Let $K_1 = (\{x, y\}, \{\mathit{person}(x), \mathit{casino}(y), \mathit{go\text{-}to}(x, y)\})$ and $K_2 = (\{z\}, \{z = \mathit{money}, \mathit{lose}(x, z)\})$. Then $f \models_M K$ iff for all $g_1 \supseteq_{\{x,y\}} f$ (always trivially true) such that $g_1 \models_M K_1$, there is a $g_2 \supseteq_{\{z\}} g_1$ such that $g_2 \models_M K_2$.

i.e. for every $g_1\colon U_D \to U_M$ such that:

1. $g_1(x) \in V_M(\mathit{person})$
2. $g_1(y) \in V_M(\mathit{casino})$
3. $(g_1(x), g_1(y)) \in V_M(\mathit{go\text{-}to})$

there is some $g_2\colon U_D \to U_M$ such that:

1. $g_2(x) = g_1(x)$ and $g_2(y) = g_1(y)$
2. $g_2(w) = g_2(x)$
3. $g_2(z) = V_M(\mathit{money})$ and $(g_2(w), g_2(z)) \in V_M(\mathit{lose})$

Question 2

Formulate the λ-DRSs for the following lexical items:

a. *every*: $\langle\langle e,t\rangle,\langle e,t\rangle,t\rangle$ $\quad$ $\lambda P \lambda R.([x \,|\, \varnothing] + P(x)) \Rightarrow R(x)$

b. *a*: $\langle\langle e,t\rangle,\langle e,t\rangle,t\rangle$ $\quad$ $\lambda P \lambda R.[x \,|\, \varnothing] + P(x) + R(x)$

c. *have*: $\langle e,\langle e,t\rangle\rangle$ $\quad$ $\lambda x \lambda y.[\varnothing \,|\, \mathit{have}(x,y)]$